

Rajalakshmi Engineering College

Name: Chandhru L
Email: 241801035@rajalakshmi.edu.in
Roll no:
Phone: 9677122982
Branch: REC
Department: AI & DS - Section 5
Batch: 2028
Degree: B.E - AI & DS

Scan to verify results



2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 10_Q3

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : COD

1. Problem Statement

Priya is analyzing encrypted messages in a research project. She wants to analyze the frequency of each character in a given paragraph. The characters should be stored in a TreeMap so that the output is sorted in ascending order of characters automatically.

You are required to build a Java program that:

Uses a `TreeMap<Character, Integer>` to count how many times each character appears in the message. Ignores spaces and considers only alphabets (case-sensitive). Outputs the frequencies of characters in sorted order.

You must use a TreeMap in the class named MessageAnalyzer.

Input Format

The first line of input contains an integer n, the number of lines in the message.

The next n lines each contain a string (the encrypted message line).

Output Format

The first line of output prints: "Character Frequency:"

Then print each character and its frequency in the format: "<character>: <count>"

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 2

Hello World

Java

Output: Character Frequency:

H: 1

J: 1

W: 1

a: 2

d: 1

e: 1

l: 3

o: 2

r: 1

v: 1

Answer

```
// You are using Java
```

```
// You are using Java
```

```
import java.util.*;
```

```
public class Main {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        int n = Integer.parseInt(sc.nextLine());  
        TreeMap<Character, Integer> t = new TreeMap<>();  
        while (n-- > 0) {  
            String s = sc.nextLine();
```

```
        for (char c : s.toCharArray()) {
            if (Character.isLetter(c)) {
                t.put(c, t.getOrDefault(c, 0) + 1);
            }
        }
        System.out.println("Character Frequency:");
        for (Map.Entry<Character, Integer> e : t.entrySet()) {
            System.out.println(e.getKey() + ": " + e.getValue());
        }
    }
}
```

Status : Correct

Marks : 10/10